CS 571

Homework 2 – Entity and Relation Extraction

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This homework is finished with the assistance of Codex.

I. Supervised Entity Tagging

Task and protocol. I treat named-entity recognition as token-level BIO classification over the six Re-DocRED types: PER, ORG, LOC, TIME, NUM, and MISC. The model is trained only on the 3,053 documents in `train_revised.json`. The 500-document dev split is used for feature and hyperparameter selection. After fixing every choice, I run the model once on the 500-document test split. Evaluation is at the mention-span level: a prediction is correct only when sentence index, start, end, and type all exactly match a gold mention. Thus every span, rather than every coreference cluster, contributes one item to micro precision, recall, and F1.

Model and features. The tagger is an averaged multiclass linear classifier trained with stochastic gradient descent and log loss. Sparse categorical features are mapped with `FeatureHasher`; this bounds memory and avoids learning a separate vocabulary. For token w_i, the representation contains:

- the original and case-folded token, compressed character shape, capped length, and indicators for uppercase, title case, lowercase, digits, embedded digits, hyphens, and alphanumeric form;
- all case-folded prefixes and suffixes of lengths 1–5;
- case-folded token and shape features from a ± 3 token window, explicit sentence-boundary markers, and left/right lexical bigrams;
- the preceding BIO tag (gold during training and predicted during decoding), which gives the otherwise local classifier sequence context.

When gold spans overlap, longer spans are assigned first and later conflicts are skipped so that each sentence has one valid BIO sequence. At inference, orphaned I-tags are repaired to B-tags. Repeated mentions with the same type and normalized surface string are grouped into one Re-DocRED entity; grouping does not affect mention-level evaluation.

Selected hyperparameters. The final classifier uses 524,288 hashed dimensions, L2 regularization with $\alpha = 10^{-6}$, at most 20 epochs, convergence tolerance 10^{-3}, averaged weights, and random seed 571. Because O tokens dominate the corpus, they receive sample weight 0.60 while entity tokens receive weight 1.00. The loss is one-vs-rest logistic loss; feature-hash signs are nonnegative. These are the defaults in `ner_predict.py`, while command-line options expose all of these settings for reproducible experiments.

Validation and parameter selection. I used a sequential, dev-only coordinate search, beginning with the original configuration and retaining changes that improved strict micro-F1. The table shows representative trials; settings not named in a row remain those of the preceding retained configuration. The final three feature changes are cumulative. No test labels were inspected during selection.

Trial	Main settings / change	Micro P	Micro R	Micro F1
Initial	$\alpha = 2e{-}6$, 15 epochs, O=.35, window=2, affix=4, hash=2^{18}	.7400	.7410	.7405
Regularization	$\alpha = 1e{-}6$, 20 epochs	.7427	.7448	.7437
Stronger regularization	$\alpha = 5e{-}6$, 20 epochs	.7324	.7331	.7328
Class balance	O weight=.25	.7355	.7360	.7358
Class balance	O weight=.45	.7487	.7509	.7498
Class balance	O weight=.60	.7503	.7533	.7518
Class balance	O weight=.80	.7484	.7506	.7495
Context	window $2 \rightarrow 3$ (O=.60)	.7528	.7542	.7535
Hash capacity	$2^{18} \rightarrow 2^{19}$	.7532	.7554	.7543
Selected	maximum affix length $4 \rightarrow 5$	**.7563**	**.7592**	**.7578**

The search shows that reducing regularization and increasing the O-token weight improve the precision–recall balance. A third context token, more hash capacity, and length-5 affixes provide smaller cumulative gains. The selected model improves dev micro-F1 by .0173 over the initial configuration.

Final span-level results. The locked model transfers from dev to test with only a .0064 decrease in micro-F1. TIME is easiest, followed by NUM; ORG and MISC remain the most difficult, mainly because of low recall and type ambiguity.

	Dev			Test		
Type	P	R	F1	P	R	F1
PER	.7696	.8091	.7889	.7596	.8176	.7875
ORG	.6995	.5740	.6306	.6682	.5535	.6054
LOC	.7421	.8457	.7905	.7558	.8342	.7930
TIME	.9328	.9205	.9266	.9226	.9033	.9128
NUM	.7963	.8627	.8281	.8104	.8839	.8456
MISC	.6019	.5086	.5514	.5754	.4750	.5204
Micro	**.7563**	**.7592**	**.7578**	**.7522**	**.7506**	**.7514**
Macro	.7570	.7534	.7527	.7486	.7446	.7441

Error analysis

For analysis only, unmatched gold and predicted spans are paired one-to-one. Exact-boundary pairs with different types are paired first; remaining overlapping pairs are matched by same type and then token intersection-over- union. On test there are 2,766 paired near misses: 1,780 type-only errors, 600 boundary-only errors, and 386 with both errors. Consequently 35.6% of paired near misses involve a boundary error and 78.3% involve a type error (the categories overlap). A further 464 gold mentions are completely missed and 435 predictions are spurious. Among the 986 boundary-involved pairs, 567 are too narrow, 412 too wide, four shifted, and three mixed.

Gold \ predicted	PER	ORG	LOC	TIME	NUM	MISC
PER	0	47	168	1	2	91
ORG	161	0	333	2	5	148
LOC	170	114	0	1	1	141
TIME	0	6	10	0	37	13
NUM	0	3	0	15	0	3
MISC	184	185	309	10	6	0

Table 1. Test type-confusion counts for paired near misses with type disagreement. Rows are gold types and columns are predicted types; the diagonal is zero by construction.

The largest directed confusion is ORG→LOC (333), followed by MISC→LOC (309). MISC also disperses into PER and ORG, which is consistent with its heterogeneous semantic role. LOC recall is high but its false positives reduce precision, while ORG exhibits the opposite recall problem. TIME/NUM confusions reflect date-like and quantity-like expressions. The high share of too-narrow spans suggests that local lexical evidence often finds an entity head but omits modifiers; stronger structured decoding or a span-level model would be a natural next improvement.